



Département d'Enseignement et Recherche en Biologie
École Normale Supérieure de Paris-Saclay

Année universitaire 2025-2026

Rapport de stage de L3

Bacterial motility and the evolution of cooperation in microbial communities

Tristan GODART

Stage encadré par Nuno OLIVEIRA

LBPA

Stage effectué du 1er juin au 31 juillet 2026

Contents

1	Introduction	1
1.1	Microbial cross-feeding	1
1.2	Bacterial motility and the evolution of cross-feeding	1
1.3	Modelling motility through cellular automata models	2
1.4	Objectives of the internship	3
2	Methods	3
2.1	Logic of the models	3
2.2	One-secretion model	4
2.3	Two-secretions model	5
3	Results	6
3.1	One-secretion model	6
3.1.1	Exploitation outweighs cooperation at high levels of motility	6
3.1.2	Chemotaxis promotes local interactions	8
3.2	Two-secretions model	10
3.2.1	Between-genotype cooperation arises at intermediate levels of motility	10
3.2.2	Chemotaxis promotes cross-feeding	12
3.2.3	Density-dependent motility hinder inter-genotype cooperation and favor intra-genotype cooperation	14
3.2.4	Passive genotypes hinder inter-genotype cooperation	16
4	Discussion	18

Abréviations

prob. Probability

1 Introduction

1.1 Microbial cross-feeding

Micro-organisms usually live in dense communities with great genetic diversity and species richness. This diversity is paramount to the larger-scale effects of these societies, such as in microbiomes, in polymicrobial infections, and in nutrient recycling. Given this diversity and the many phenotypes that influence the reproduction and survival of surrounding cells [1], ecological interactions are thought to be both common and strong in microbial communities. Examples of such traits include the production of a broad range of secretions, such as extracellular enzymes and scavenging molecules, carbon sources, detoxification agents, or amino acids [2].

These "leaky" traits are often metabolically costly to produce [2], which hints towards a cooperative "altruistic" behavior where an actor disadvantages itself to benefit others. As for the evolution of altruism in animals, a central explanation for such cooperative phenotypes in microbes is kin selection: microbes help cells of related genotypes at the cost of their personal fitness, but thereby raising their *inclusive fitness*, namely their indirect contribution to the gene pool through the reproduction of related cells [3]. Kin selection can readily evolve if it satisfies *Hamilton's rule* :

$$rB > C, \quad (1)$$

where $r \in [0, 1]$ is the genetic relatedness of both cells, $B \geq 0$ is the additional reproductive benefit for the recipient cell and $C \geq 0$ is the reproductive cost to the actor cell [3].

However, microbes also commonly interact with cells of different genotypes (i.e. $r \ll 1$) in complex ecological networks [4], which is not explained by kin selection as it does not satisfy Hamilton's rule. These *cross-feeding* relationships, where cells of different genotypes exchange costly secretions, is often described as cooperation. Cross-feeding can be obligate, meaning that some genotypes cannot grow in the absence of other genotypes, which may explain why most microbial species cannot grow in the laboratory when in monoculture [2]. Additionally, a growing number of arguments such as genetic drift, efficiency of metabolic fluxes, intermediate privatization of resources, and the Black Queen hypothesis posit that obligate cross-feeding should commonly evolve in bacterial communities [2, 5, 6].

One microbial interaction that has been explored is *syntrophy*, where one species feeds on the toxic waste byproduct of another species. Although syntrophy is likely to be important in microbial communities, it is fundamentally different from cross-feeding since the underlying metabolic cost is also very low ($r \ll 1, C \approx 0$), which makes Hamilton's rule (1) inapplicable.

1.2 Bacterial motility and the evolution of cross-feeding

Motility, namely the ability to move independently using metabolic energy, is an prevalent characteristic of bacteria. Although its underlying molecular mechanisms can be described in rich details [7], its ecological impacts are less explored.

Motility is expected to be a strong ecological modulator, through colonization and the modulation

of ecological interactions, including both mutualism and competition.

Bacterial motility is diverse in natural ecosystems. For example, bacteria can detect concentration gradients of chemicals and swim upwards or downwards it, in a process called chemotaxis [8]. Their speed can also be modified in response to other factors such as the local density of cells. Both of these examples will be considered in this report.

The first eco-evolutionary model used to understand cross-feeding evolution [9] showed that genetic mixing has a strong impact on the potential for the evolution of cross-feeding relationships. Overall, their model showed that cross-feeding is limited by natural selection, and fragile and inefficient when it is selected. This result is challenged by the prevalence of cross-feeding relationships in natural ecosystems [2, 4], which hints that their model missed a key characteristic thereof. To understand what has been missed, genetic mixing must be modeled explicitly through the motility of the bacteria.

1.3 Modelling motility through cellular automata models

In this report, the problem of the emergence of obligate cross-feeding in motile populations is tackled through *evolutionary game theory*, a theoretical framework applying the tools of game theory to evolving populations where the success of each genotype relies on the others. In this context, *cellular automata* models, where cells live in a discrete lattice in two dimensions, have emerged as a paradigm to describe interactions between several genotypes [10, 11]. Keeping track of individuals, their local environments and behaviors allows for a detailed representation of the microbes in a spatial context [12], which can reveal complex dynamics emerging from individual-level rules [11, 12] such as *spatial structures*, i.e. differences of local conditions experienced by the microbes [12].

Spatial structures are paramount for modeling cross-feeding as they are inherently linked to ecological interactions. Interactions are shaped by spatial structures as bacteria can only interact with their closest neighbors [12]. Inversely, ecological interactions shape the spatial structures through an emerging property called *self-organization*, caused by the fact that cells grow faster when in mutualistic interactions and slower when in competition [12].

Spatial structures have been shown to disfavor cross-feeding [9], but the model used did not include motility. In cellular automata models, motility has been shown to go against self-organization and to disrupt the formation of spatial structures, even removing them entirely and making the population homogeneous if powerful enough [11]. This suggests that motility could alleviate the constraints observed in [9] and enable cross-feeding evolution.

Such model can be *stochastic*, meaning that reactions happen randomly and in a random order, according to a set probability distribution. This is particularly relevant when modeling motility at a single-cell level, as bacterial motion is inherently stochastic. Moreover, the addition of a random noise in the behavior of each individual allows for the emergence of heterogeneity from individual-level rules, which can give rise to rich emerging properties at the scale of the population.

1.4 Objectives of the internship

The purpose of this internship was to study how motility can affect the evolution of cross-feeding in bacterial populations. This was done by first considering unbiased motility, and then by adding rules such as chemotaxis or density-dependent motility to better capture bacterial behavior in natural ecosystems.

To this end, a stochastic cellular automata model based on the well-established spatial rock-paper-scissors model [11] was used. This model was extended to allow cells to invest in multiple distinct cooperative secretions, so that there is potential for different genotypes to exchange secretions with each other, modeling cooperation between different genotypes [5, 6, 9]. Analysis of these models shows that between-genotype cooperation arises at intermediate levels of motility and is favored by chemotaxis, but mostly disfavored by motility responses to the local density of cells and highly disfavored by passive genotypes which prevent mixing between genotypes.

2 Methods

The goal of this internship was to model how motility shapes the potential for cooperation in bacterial communities with leaky functions. These common goods can be pivotal in microbial growth, both in natural and disease settings. Two models are introduced in this report. The first is a producer-defector model, aiming to assess how motility affects the exploitation of productive genotypes by unproductive ones. The second introduces partial secretors, allowing cells of different genotypes to exchange secretions and aiming to extend the first model to feature cooperation between genotypes.

2.1 Logic of the models

The way common goods influence growth is through their considerable metabolic cost for their producers [13], and through the reproductive benefit to all cells in a producer's vicinity [1]. To reflect both mechanisms, we use an agent-based spatial model where the fitness of each individual is directly linked to secretion production and availability. This single-cell approach compels us to explicitly take the stochastic character of the interactions between cells into account, as an inherent nature of real ecosystems at this scale. In the context of competitions between different genotypes, such stochastic spatial systems have been shown to allow for the emergence of dynamic spatial structures when considering the motility of agents [11].

Although the analysis will be phrased in terms of secretions like enzymes that liberate nutrients, it also captures a diverse range of leaky beneficial traits, such as detoxifying enzymes like catalase or amino acids. To reflect this diverse array of common goods, secretions can be set to either be essential for bacterial growth or not, meaning that secretions can be set to either accelerate growth, or to enable it altogether.

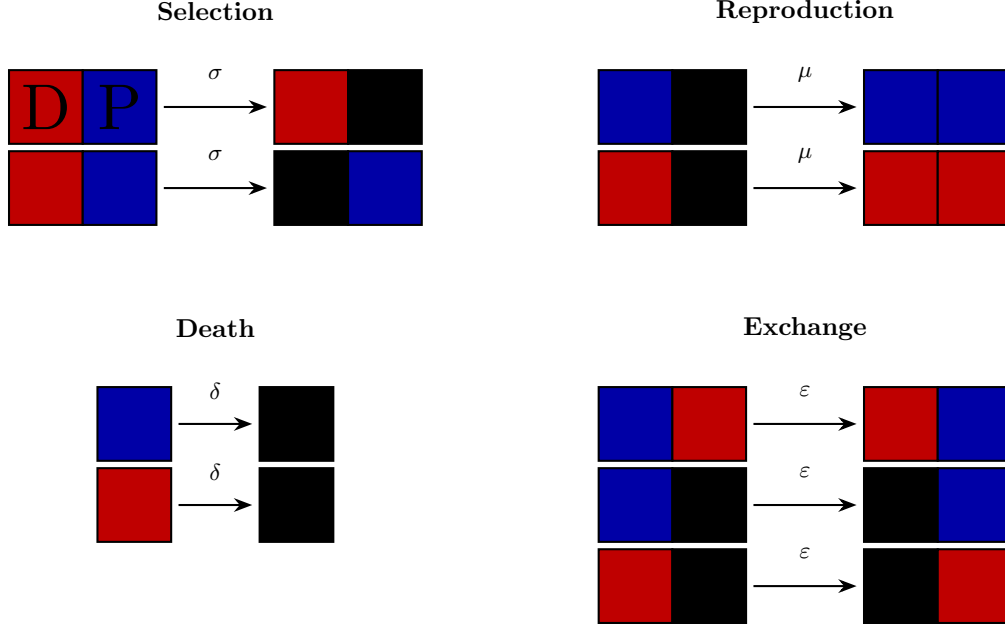


Figure 1: The rules of the stochastic model. Individuals of two competing strains [1] (blue) and [0] (red) occupy sites of a lattice. They interact with one of their 4 nearest neighbors through reactions of reproduction, selection or death, which occur as Poisson processes at rates μ , σ and δ respectively. Selection reflects the local effectiveness of each strategy: bacteria can outcompete bacteria of other strains with lower fitness, which kills them and replaces them with an empty site (black) (Algorithm 1). Reproductions of individuals are only allowed on empty sites to mimic a finite carrying capacity of the system, and their success is dependent on their fitness (Algorithm 2). Death happen without respect to fitness or strain, to mimic a finite lifespan of individuals and to make successful reproduction events primordial in the long-term longevity of the strain (Algorithm 3). Bacteria are also endowed with motility: at exchange rate ϵ , they swap positions with an adjacent space.

2.2 One-secretion model

Consider motile cells of two genotypes: producers of a given secretion, referred as [1], and defectors [0] who don't produce it. These cells are arranged on a spatial lattice with periodic boundaries, where they can only interact with their 4 nearest neighbors. For the possible interactions, a variation of the stochastic cellular automata model established in [11] is considered. This framework can model the interplay between local interactions and motility, and has been shown to feature a direct link between motility and emerging spatial structures [11].

Schematic illustrations of the model's dynamics are provided in Figure 1. The basic reactions comprise reproduction, selection and death processes, which occur at rates μ , σ and δ respectively. Cells' motility stems from the possibility that an individual will swap its position with an adjacent space at rate ϵ . Hence, individuals randomly migrate on the lattice. The interplay of migration with the reproduction, selection and death processes sensitively determines which evolutionary strategy will eventually dominate.

The lattice is initialized with a population of low density: on average $\frac{\#G}{50}$ random spaces are initially occupied by a random bacteria, with $\#G$ being the total number of genotypes in the lattice. Simulations steps are then performed until only one strain remains, and consist in the choice of a random individual and the choice of an interaction that it will undergo, with a random adjacent space or bacteria (in the case of reproduction and selection) or alone (in the case of death). The kinds of reaction, as well as the interplays between reactions and exchanges, are computed efficiently using

Gillespie's algorithm [14]. The unit of time of the model is defined as one *generation*, when every individual has reacted on average once. The reproduction, selection and death rates are considered constant, with $1 = \mu = \sigma = 20\delta$, so that bacteria can reproduce without secretions in the case of nonessential secretions.

The local success of each evolutionary strategy is modeled through a fitness score, which can be seen as the potential reproductive success of each cell. Fitness depends both on the local concentration of secretions and on the eventual metabolic cost of producing secretions:

$$f = b + ng - sc, \quad (2)$$

where $b \geq 0$ is the basal fitness that every individual has, without respect to secretions. n is the number of producer [1] in a neighborhood of 5×5 spaces centered around the bacteria, and g is the fitness gained from the secretions produced by one of them. This way, the term ng accounts for the total fitness gained for all secretors in the vicinity of the bacteria, including itself. Similarly, c is the metabolic cost of secretions, and $s \in \{0, 1\}$ denotes the number of secretions produced by a bacteria, so that the metabolic cost of secretions is only applied to producers. If $g = c$, the reproductive benefit of secretions compensates exactly the metabolic cost induced to the secretor. This way, the producers are not self-sufficient and the model captures intra-genotype cooperation instead of self-sufficiency.

In this report, each evolutionary strategy will be compared through the *extinction probability* of their corresponding strains, which is defined here as the frequency of simulations where the strains eventually goes extinct. This approach requires the simulations to run until one strain goes extinct. Past that point, the simulation then runs for a fixed amount of time to check whether the surviving strain can survive alone. For example, if $b = 0$, the strain [0] cannot survive on its own once the strain [1] is extinct, so it will go extinct as well.

2.3 Two-secretions model

The 1-secretion model allows for the modeling of cooperative exchanges in bacterial populations, but fails to model this cooperation between different genotypes. This is the purpose of the 2-secretions model, which uses intermediate complementary genotypes to study obligate cross-feeding, similarly to other models in the literature [5, 6, 9].

The 2-secretions model has four genotypes, labeled according to whether they produce each secretion [9]. These four genotypes are a producer of both secretions [1, 1], a producer of the first trait only [1, 0], a producer of the second trait only [0, 1] and a non producer [0, 0].

Before the first generation occurs, bacteria of these four genotypes are randomly placed on the lattice so that $\frac{4}{50}$ spaces on average are filled with bacteria. This choice of beginning the simulations with each genotype being present with equal, low densities is made to capture two related scenarios [9]. The first scenario corresponds to a single strain having lost the ability to produce secretions, and the second scenario corresponds to the strains coming from different species of bacteria which fill similar ecological niches.

The fitness formula now takes into account both secretions :

$$f = b + [\omega \max(n_1, n_2) + (1 - \omega) \min(n_1, n_2)]g - sc. \quad (3)$$

Here, n_1 and n_2 denote the number of producers of each secretion in a 5×5 space around the bacteria. The 2-secretions model introduces a new parameter ω , depicting the redundancy of each secretion. If $\omega = \frac{1}{2}$, then both secretions can raise the fitness independently. If $\omega = 0$ however, both secretions are necessary for a bacteria's fitness to increase, which simulates a scenario where both public goods are required simultaneously for a benefit to be observed. Together with $b = 0$, this allows for the modeling of obligate cross-feeding, where both partners cannot grow in the absence of the other [2].

Once again, the equality $g = 2c$ is maintained thorough the study of the 2-secretions model, so that producers are not self-sufficient. This will allow the model to compare intra-genotype cooperation and inter-genotype cooperation, without having one dominate because of the self-sufficiency of its cells.

3 Results

3.1 One-secretion model

3.1.1 Exploitation outweighs cooperation at high levels of motility

The first step is to study how different motility rates ε affect the outcome of the competition between producers and defectors, when each bacteria move randomly. Figure 2a gives the extinction probabilities as a function of the exchange rates ε . For both unessential and essential secretions, cooperation is favored at low motility, whereas high motility favors exploitation by defectors. This result correspond with well-known cooperator-cheater dynamics, where limited genotype mixing leads to the evolution of secretion, whereas high genotype mixing leads to the loss of secretions [9, 15, 16]. Interestingly, the outcomes of the simulations with essential secretions are a lot less deterministic than with unessential secretions. This can be explained by the lower fitness of bacteria with essential secretions, which lower the probability of reproduction events to be successful (Figure 2e). In turn, less cells are present on the lattice, which decreases the number of reactions and exchanges and make the random outcomes of those that remain more prevalent (Figure 2f).

Figure 2b shows that the outcome of the competition also affects the community's productivity: the highest final density of cells is attained when producers are victorious in essential secretions, where approximately 95% of the lattice is occupied by a cell. Why this value in particular ? If only one strain remains, spatial structures disappear and the system can be approximated as well-mixed. With these assumptions in mind, the steady-state number of cells in the system can be approximated by the following logistic equation:

$$N'(t) = \bar{f}\mu N(t) \left(1 - \frac{N(t)}{L^2}\right) - \delta N(t), \quad (4)$$

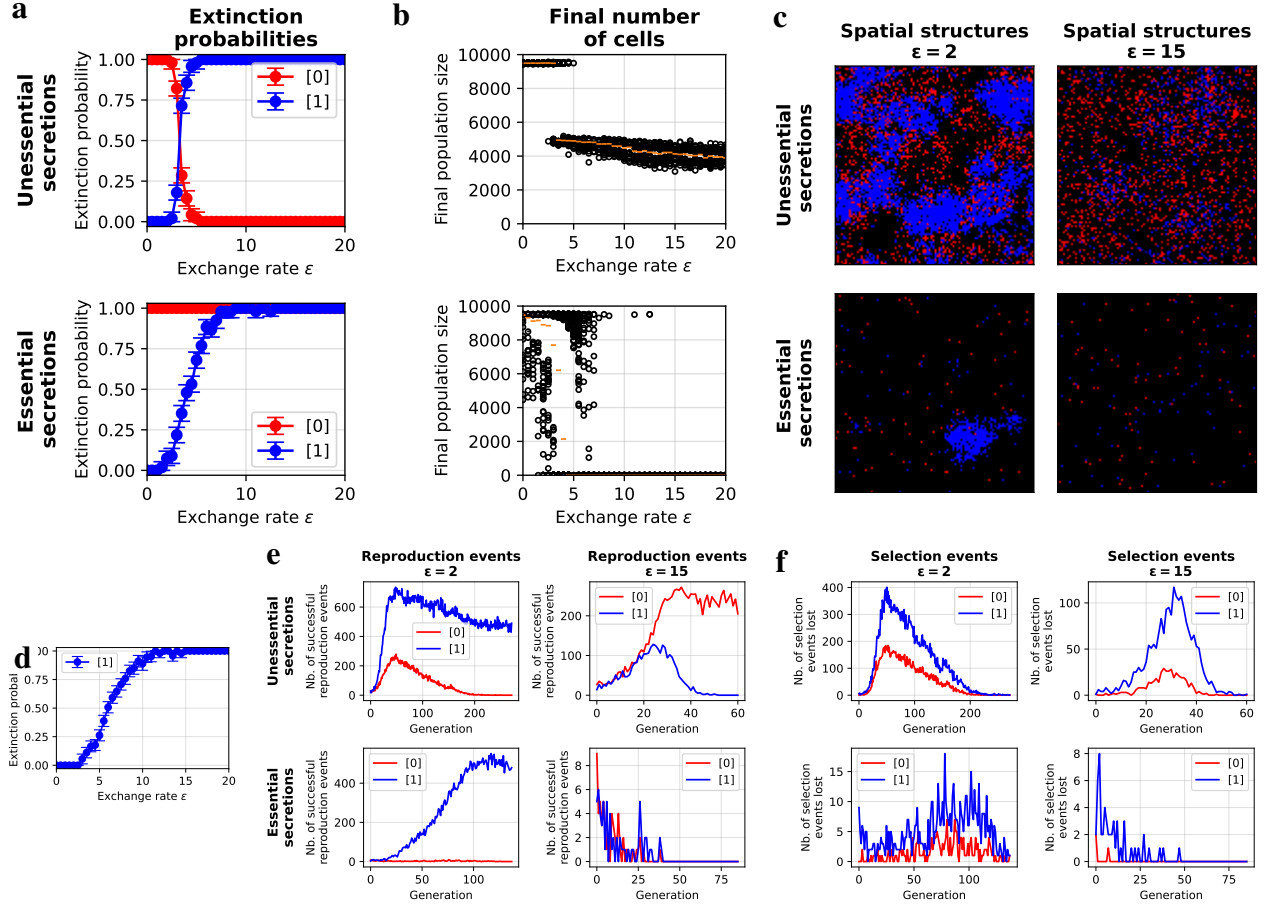


Figure 2: Dynamics of the 1-secretion model under unbiased motility. **a** Extinction probabilities of each strain for unbiased motility in the 1-secretion model, with unessential and essential secretions. If secretions are unessential, $b = 0.1$ in the fitness formula (2), whereas $b = 0$ if secretions are essential. This change makes bacteria unable to reproduce without being near a producer [1]. **b** Number of living cells for both unessential and essential secretions, recorded when one genotype goes extinct. The median is marked with an orange line. **c** Snapshots of spatial structures for both unessential and essential secretions, taken with the same exchange rate and at the same generation for each column. Producers [1] are represented with blue pixels, defectors [0] with red pixel, and the black pixels are empty space. **d** Extinction probability of producers in essential secretions, when simulated without defectors. **e** Number of successful reproduction events for each strain at each generation, in the simulation used to generate Figure 2c. **f** Number of death to selection events for each strain at each generation, in the simulation used to generate Figure 2c.

where $\bar{f} \in [0, 1]$ is the average fitness of cells, and L is the size of the square lattice, so that L^2 is the total number of sites and corresponds to the load capacity of the lattice, or to the maximal population size. With the previous assumptions, $\bar{f}N(t)/L^2$ is the probability that ones reproduction event is successful. For a population of producers, $\bar{f} = 1$ and the solution of (4) converges towards $0.95L^2$, which explains the final density observed before. For a population of defectors and with unessential secretions, $\bar{f} = b = 0.1$ and the total number of cells converges towards $0.5L^2$.

However, these values do not always correspond to what is displayed on Figure 2b. This figure shows the number of cells when the competition is resolved, i.e. when the last cell of a strain dies, leaving the whole lattice for the other strain. For example, at high levels of motility and with unessential secretions, the number of cells when the last producer dies is lower than $0.5L^2$, which means that the defectors do not win because they exploit productive spatial structures, but because they prevent the formation of such structures in the first place. Indeed, if the former case were true, Figure 2b would show final densities larger than $0.5L^2$. On the contrary, at lower levels of motility, the number

of cells when the last defector dies is close to $0.95L^2$, which is the steady density according to (4). This means that a producer victory is slower, requiring steady state to be attained before the last defector dies. The slowness of this victory is consistent with the fact that producers lose most selection events.

Figure 2b shows qualitatively different final densities when secretions become essential. Final densities of $0.95L^2$ can still be seen, corresponding to cases where producers won after having reached steady state. Final densities of 0 can also be seen, which correspond to cases where the whole community goes extinct. Figure 2d displays extinction probabilities when producers are simulated without defectors, and shows that the extinction of the whole community is directly caused by the fact that producers cannot form long-term spatial structures with essential secretions at high levels of motility, which makes them die more often than they reproduce. Adding defectors accelerates this effect, but do not change qualitatively the extinction probabilities (Figure 2a). This scenario, where a bacterial population goes extinct because a common good it produces escapes it, has been considered in the scientific literature [16], but through the medium's viscosity and not through bacterial motility.

Final densities between $0.95L^2$ and 0 can also be seen for essential secretions in Figure 2b, showing that the competition can be resolved far before the population has reached its steady state. This is not the case with unessential secretions, meaning that their essentiality accelerates the competition.

But how exactly does motility affect the population? This can be explained by the spatial structures observed in Figure 2c. With low exchange rates, nearby producers [1] have many opportunities to reproduce (Figure 2e). A fraction of these opportunities generate new producers, which increases the fitness of nearby bacteria and the probability of later reproduction events to be successful. Over generations, this positive feedback loop leads to the formation of clusters of producers, which outnumber the defectors [0]. Note that the producers usually lose the selection events facing them with defectors (Figure 2f), but globally outcompete the defectors because of the higher density cells inside their cluster (Figure 2e).

On the other hand, with a high exchange rate, each bacteria moves several times before having a single opportunity to reproduce. Even after a successful reproduction event, both daughters cells move apart before the opportunity to reproduce arises again, which reduces their fitness. Over generations, producers are unlikely to form tight clusters, and the spatial structures observed at low motility are lost. With unessential secretions, producers are more sparse, which facilitates their exploitation by defectors. With essential secretions, death happens more often than reproduction, making the community collapse immediately (Figure 2d).

3.1.2 Chemotaxis promotes local interactions

What happens when the motility is not random, but targeted towards the secretions ? In real ecosystems, chemotaxis promotes interactions between individuals [8], which could help bacteria of our models in surviving the harsher conditions. In the 1-secretion model, chemotaxis is implemented according to Algorithm 6. Before moving, the number of producers in areas of 7×7 spaces centered in each space where the bacteria could move is measured. This area is a lot larger than the one used to compute the fitness in (2), to reflect how bacteria can detect diminutive concentrations of chemi-

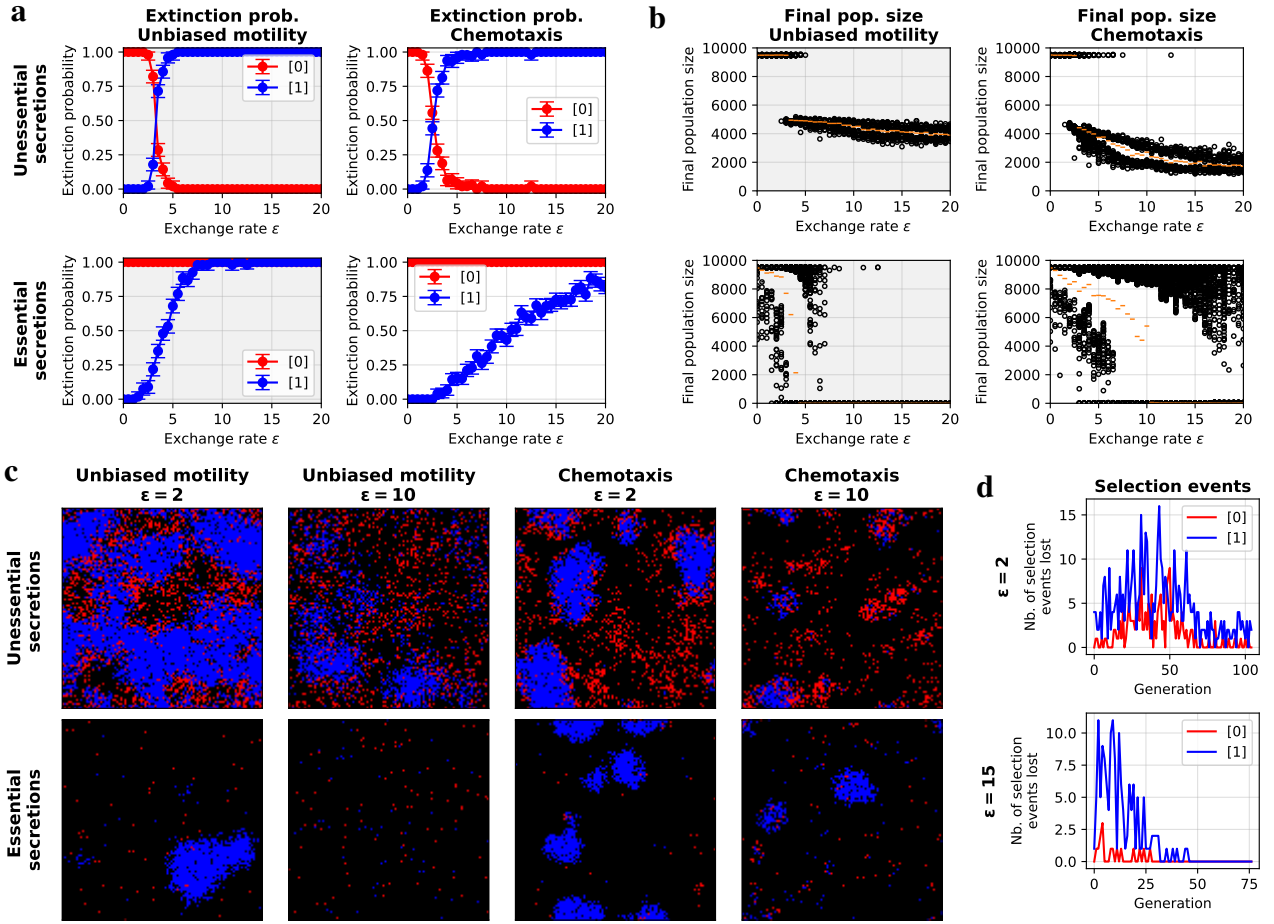


Figure 3: Dynamics of the 1-secretion model with chemotaxis. **a** Extinction probabilities with unbiased and biased motility, and example of spatial structures under biased motility. The grayed column depicts data which has already been shown in this report and which should be taken by the reader as controls for this figure. **b** Number of living cells for both unessential and essential secretions, recorded when one genotype goes extinct. The median is marked with an orange line. **c** Example of spatial structures with unbiased motility or chemotaxis, for both unessential and essential secretions. The snapshots of the spatial structures have been taken at the same generation between columns with a shared exchange rate. **d** Number of death to selection events for each strain for each generation, in the simulations with essential secretions and chemotaxis used to generate panels of Figure 3c.

cals [8]. The bacteria then still moves in a random direction, but is more likely to move in directions where more producers were detected.

This biased motility regime is, just as motility itself, a double-edged sword: it can help producers [1] in forming spatial structures, but it also can help defectors [0] in invading and exploiting them. One emergent property of this new rule is that it "glues" cells together, because cells never move away from producers. This effect is amplified when the exchange rate ϵ increases, as bacteria of both strains can explore more space and have therefore more opportunities to detect secretions.

The outcome of the competition in the 1-secretion model with chemotaxis can be seen in Figure 3a. Chemotaxis favors defectors when secretions are not essential, but favor producers when secretions are essential.

How can chemotaxis have different effects ? When secretions are not essential, Figure 3b shows that chemotaxis accelerates the simulation, with defectors winning further away from their steady state described in (4) than with unbiased motility. This correlates with the corresponding spatial structures (Figure 3c), where defectors are organized in patches around the producers. This change of

behavior retrieves well-known effects of chemotaxis on bacterial populations: it facilitates local interactions [8]. In this case, the prevalent interactions are selection events, which are won by defectors in most cases (Figure 2f).

For essential secretions, the previous section showed that the fate of the population was not shaped by the defectors' ability to invade spatial structures, but by the producers' ability to build one (Figure 2d). This means that the prevalent interaction in this case is reproduction rather than selection, which explains why chemotaxis favors producers. As it can be seen in Figure 3c, chemotaxis promotes the formation of spatial structures when secretions are essential.

But why does the population go extinct at high motility rates ? Simulating producers without defectors, similarly to what has been done in Figure 2d, shows that they never go extinct, regardless of their exchange rate (data not shown). In fact, producers are advantaged by a higher exchange rate in this case as it offers more opportunity to find each other and then to stick together via chemotaxis. This time, their extinction comes from their exploitation by defectors, which can find them more easily at high exchanges rates ϵ . Indeed, Figure 3d shows that producers lose more selection events in the first generations when the motility is increased.

3.2 Two-secretions model

3.2.1 Between-genotype cooperation arises at intermediate levels of motility

Now that the 1-secretion model allowed for a basic understanding of the ecological effects of bacterial motility, a more complex model can be introduced. The 1-secretions model is extended to model cross-feeding similarly to what has been done in the literature [9]: the lattice is still inhabited by producers $[1, 1]$ who produce all secretions, and defectors $[0, 0]$ who don't produce anything. However, this model introduces intermediate cross-feeding genotypes $[0, 1]$ and $[1, 0]$ who produce one secretion each, with intermediate metabolic cost and reproductive benefits compared to producers.

How does motility affect cross-feeding ? Figure 4a shows that inter-genotypes cooperation is selected at intermediate motility. This result is similar to what has been observed before in another model, as cross-feeding has been shown to be selected by intermediate levels of genetic mixing [9]. Intuitively, cross-feeders need motility to find their metabolic partners, but too much motility is harmful for them as it makes it harder to build spatial structures (similarly to the way that high motility affect producers in Figure 2c). If secretions are not essential for growth, this explains both how they are outcompeted by producers $[1, 1]$ at low motility rates and by defectors $[0, 0]$ at high motility rates. Producers $[1, 1]$ are shown to never go extinct with essential secretions when they are simulated alone (Figure 4d), which was not the case in the 1-secretion model (Figure 2d). This is because the additional secretion allow them to increase their fitness faster when they are in proximity, which allows them to build spatial structures consistently.

Interestingly, the range of motility where cross-feeders dominate the system is fixed for unessential secretions, but depends on the size of the lattice when secretions are essential for growth, whereas changing the lattice size did not provide any qualitative change in the 1-secretion model (data not shown). Why ? In the spatial structures with intermediate levels of motility (Figure 4b), cross-feeders

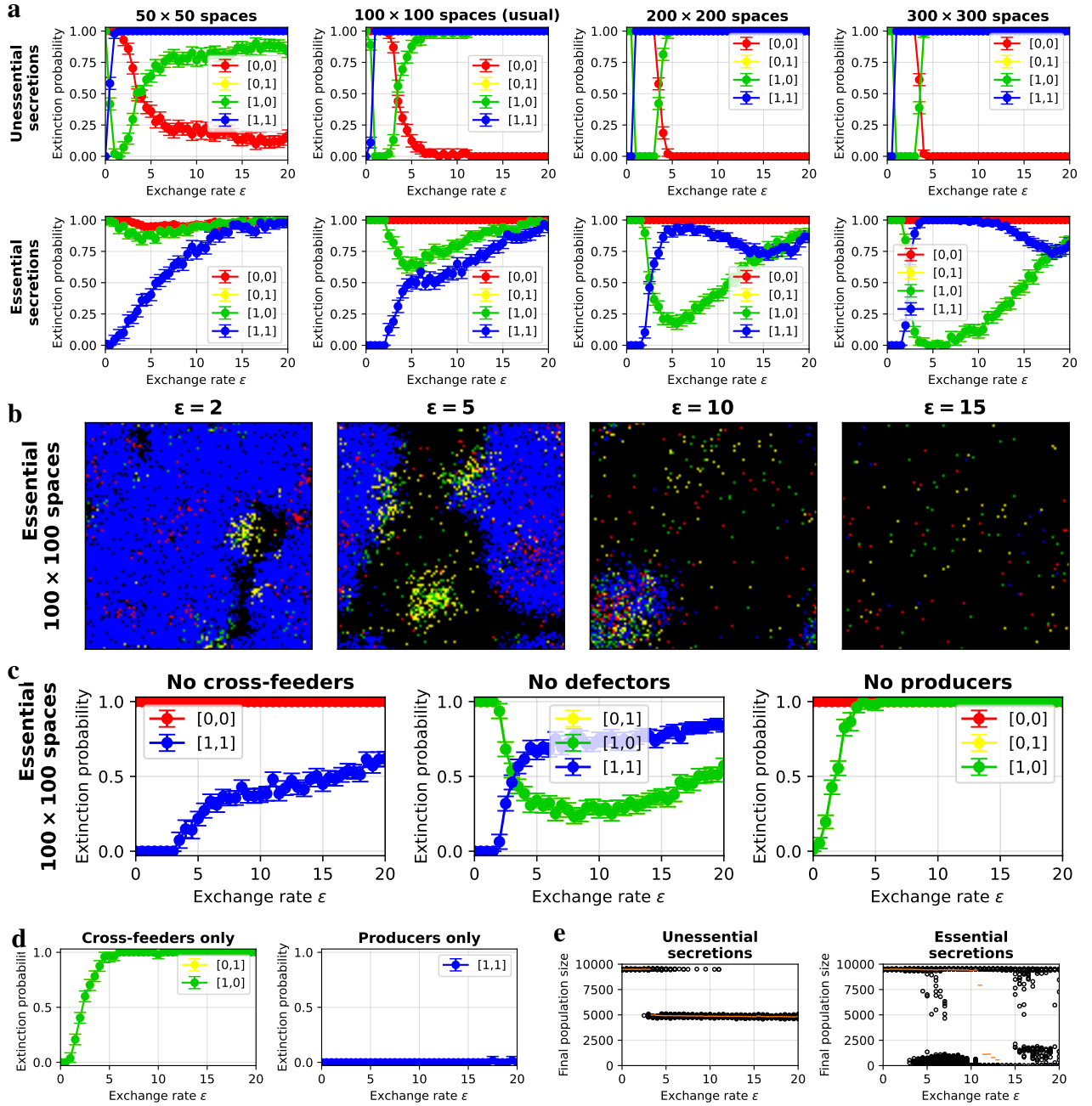


Figure 4: Dynamics of the 2-secretions model under unbiased motility. **a** Extinction probabilities of each strain as a function of the exchange rate, for both unessential and essential secretions and for several lattice sizes. **b** Examples of spatial structures, taken at generation 45 for several exchange rates, for essential secretions and the usual lattice size. **c** Pairwise interactions: extinction probabilities of each strain as a function of the exchange rate for essential secretions and with the usual lattice size, but one strain was removed from the simulation before the first generation. **d** Extinction probability of the one strain as a function of the exchange rate with essential secretions, when simulated alone. **e** Number of living cells in both unessential and essential secretions, recorded at the generation when two strains among the three went extinct. The median is marked with an orange line.

can be seen forming small clusters when secretions are essential, whereas the producers $[1, 1]$ grow abundantly. Additionally, if the producers $[1, 1]$ are removed from the simulations, the cross-feeders survive only at very low motility (Figure 4c), even when the defectors $[0, 0]$ are also removed (Figure 4d). It can be deduced that with essential secretions, the cross-feeders rely on two key factors to dominate the competition: the formation of a small spatial structure where cross-feeders can survive on their own for a small amount of time, and a contact between this spatial structure and a producers' spatial structure, which the cross-feeders will be able to exploit thanks to the halved metabolic cost of their secretions. Since changing the lattice size does not affect the initial density, a larger lattice means that more random events (such as the relative initial positions of each cell, or the outcome of each reactions) can be explored within the population. In turn, a larger lattice means that a given spatial structure is more likely to appear somewhere, which means that the first factor (small spatial structure of cross-feeders) is favored. The second factor (contact with a cluster of producers) cannot happen when there are no producers in the population (Figures 4c, 4d).

Although defectors $[0, 0]$ do not seem to have a considerable effect on the outcome, removing them from the lattice favors the cross-feeders (Figure 4c), which indicates that they can prevent one of the two events from happening. Simulating cross-feeders alone, where only the first event can happen (Figure 4d) shows little difference than when cross-feeders are simulated with defectors (Figure 4c). However, defectors are shown to have a strong effect on the producers' survival at intermediate to high motility rates (Figures 4c, 4d), so they can prevent the second event from happening. Ultimately, defectors have a negative effect on cross-feeders because they can exploit the producers before they can build their spatial structure, which the cross-feeders need to grow.

Finally, what does the community look like when the competition is resolved ? If secretions are nonessential, final densities of $0.95L^2$ and $0.5L^2$ only are retrieved. According to (4), this means that each strain has achieved its steady-state population at the moment of their victory ($0.95L^2$ for the productive strains, and $0.5L^2$ for defectors), and that each victory is slow and implies the clash of different strains organized in spatial structures. With essential secretions, final densities between 0 and $0.95L^2$ can be seen, meaning that victory can be faster and can happen before spatial structures have covered the whole lattice.

3.2.2 Chemotaxis promotes cross-feeding

The increased complexity of the 2-secretions model calls for alternative chemotaxis rules to understand its effects on the population. Three rules are studied in this section: *Chemotaxis towards what the genotype does not produce*, *Chemotaxis towards what the genotype does produce* and *Chemotaxis towards all secretions*.

Figure 5a show the results of the simulations with the three chemotaxis rules. With unessential secretions, the only rule that shows major changes compared to unbiased motility is "Chemotaxis towards what the genotype does produce", where the cross-feeders dominate at exchange rates where the defectors dominated, and can be explained by the defectors' reduced ability to exploit when they are the only ones to move randomly. However, essential secretions show consequential changes between unbiased motility and rules of chemotaxis, with cross-feeders dominating for broad ranges of

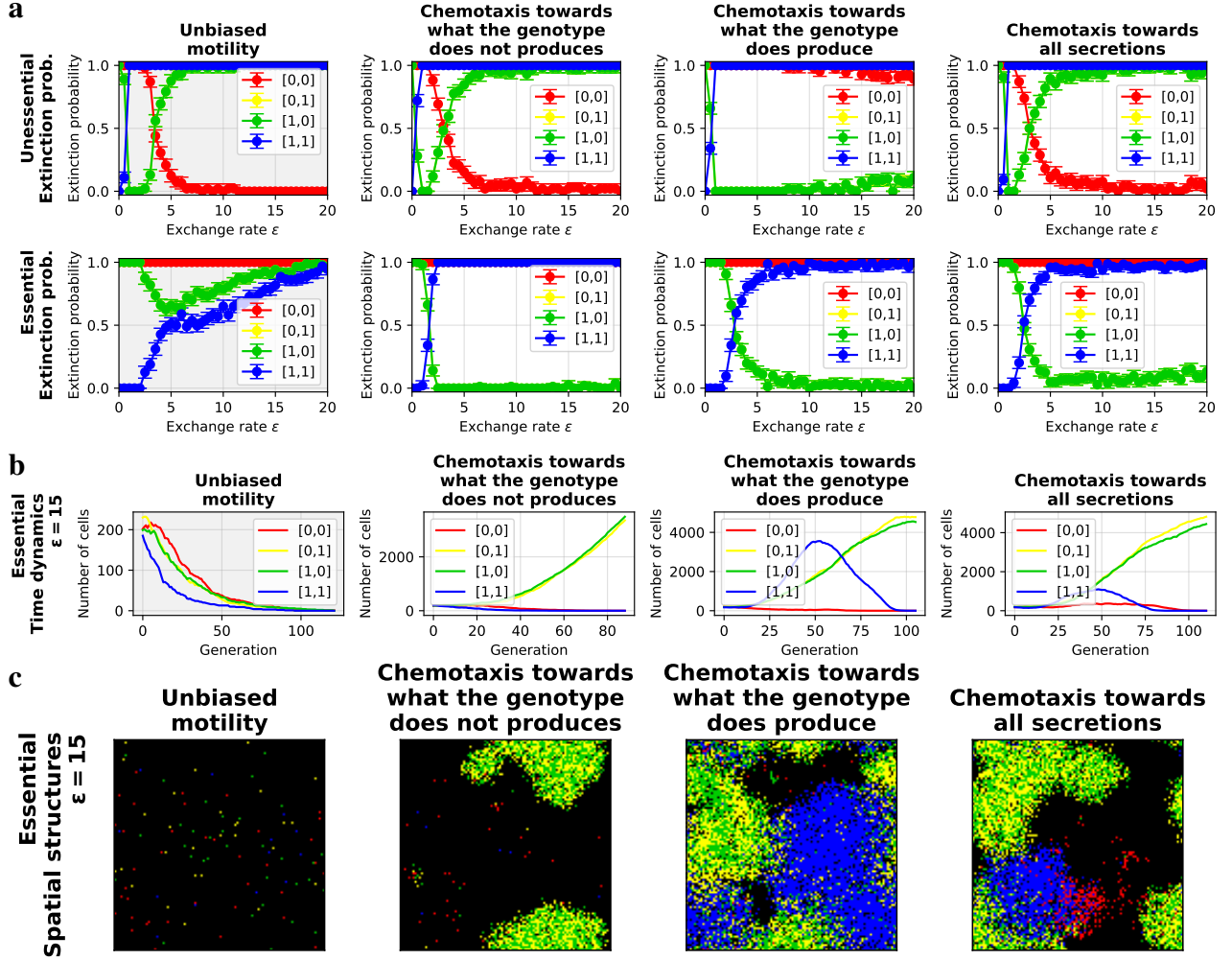


Figure 5: Dynamics of the 2-secretions model with chemotaxis. **a** Extinction probabilities of each genotype as a function of the exchange rate, for both unessential and essential secretions, and with unbiased motility and several rules of chemotaxis. The grayed panels display data that has been shown before in this report, and which should be used by the reader as controls for this figure. **b** Examples of temporal dynamics of the 2-secretion model with essential secretions, for unbiased motility and different rules of chemotaxis. Each panel display how many cells of each genotype were alive at the end of each generation. A exchange rate of $\epsilon = 15$ was used for each panel. **c** Examples of spatial structures in the simulations used to generate the temporal plots of Figure 5b. Each snapshot was taken at generation 50.

exchange rates.

How exactly do the cross-feeders dominate ? The answer is straightforward with the first chemotaxis rule: the cross-feeders are more likely to find their metabolic partner and to be "glued" to them, which makes them form spatial structures fast. Even inside the spatial structure, the cross-feeders will be attracted to areas where their metabolic partners are the most abundant, which optimizes their growth. The spatial structure can then grow without needing producers [1, 1], who still move randomly and fail to build spatial structures, thereby offering little competition (Figure 5c).

The cross-feeders' dominance with the second chemotaxis rule is more counter-intuitive: this rule disfavors mixing between genotypes, so the cross-feeders are expected to be disadvantaged when compared with unbiased motility. However, Figure 5b shows that the cross-feeders are able to survive (by building two nearby small homogeneous clusters) until the producers have built a large population, and then exploit them to grow. Once the last producer dies, the cross-feeders can survive on their own through self-organization.

The cross-feeders' also dominate at high exchange rates with the third chemotaxis rule, but their survival probability lower than with both previous rules. This is because both previous rules had a drawback for them, which is amplified when they are combined. The first rule amplifies the defectors' exploitation, but makes the cross-feeders' mixing very optimized and practically removes the producers from the simulation (Figures 5b, 5c). The second rule practically removes the defectors from the simulation (which is more visible with unessential secretions), but makes the cross-feeders mixing a lot less effective. With the third rule, both of the producers' and defectors' behaviors are optimized for growth, and the cross-feeders' mixing is less effective than with the first chemotaxis rule.

All in all, chemotaxis promotes cross-feeding as opposed to cheating in very high motility regimes, and can make the bacterial population sustainable when their motility is very high.

3.2.3 Density-dependent motility hinder inter-genotype cooperation and favor intra-genotype cooperation

Behaviors other than chemotaxis have been observed in motile bacteria. One example of such behavior is bacteria slowing down in crowded areas, which is often called "motility-induced phase separation", but will be called *density-dependent motility* in this report. This phenomenon is well-studied, and has been known to facilitate the formation of dense motionless aggregates which can be precursors of biofilms [17]. In the context of this internship, density-dependent motility is, like chemotaxis, expected to facilitate the emergence of clusters, which could favor cross-feeding.

Density-dependent motility is implemented in the 2-secretions model according to Algorithm 7. Bacteria move according to the usual exchange rate ϵ , but if the local density of cells in a 7×7 square centered around the bacteria is above a given threshold, the effective exchange rate of this bacteria will be reduced to $\epsilon_{\min} = 0.1$. The detection area is large, to reflect how bacteria can detect diminutive concentrations of signaling molecules, and to be consistent with the implementation of chemotaxis.

The effects of density-dependent motility on the 2-secretions model is summed up in Figure 6a. If secretions are not essential, the cross-feeders now do not survive at intermediate exchange rates, and producers can completely dominate the system if the threshold is low enough. If secretions

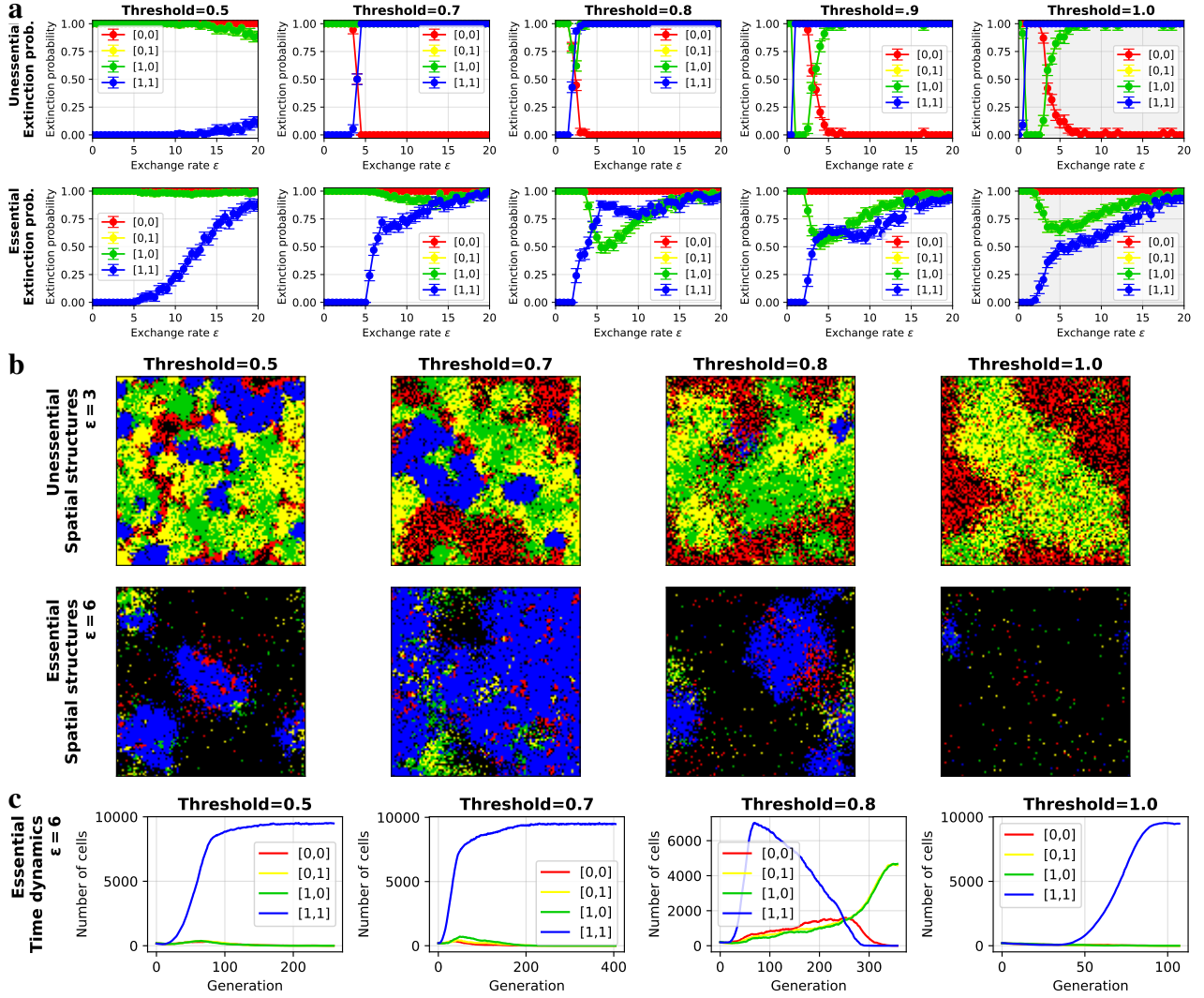


Figure 6: Dynamics of the 2-secretions model with density-dependent motility. In all subfigures, the parameter $\epsilon_{\min} = 0.1$ was used. **a** Extinction probability of each genotype as a function of the exchange rate, for both unessential and essential secretions, with several thresholds. The grayed column displays data with a threshold too high, so that the bacteria never slow down. The data is then identical to what is shown in Figure 4a. **b** Examples of spatial structures for both unessential and essential secretions and for each threshold. The snapshots were taken at generation 50 for unessential secretions, and generation 40 for essential secretions. **c** Examples of temporal dynamics of the population, with essential secretions. The simulations used to plot these panels are the same that were used in Figure 6b.

are essential, low thresholds favor producers and intermediate thresholds can favor cross-feeders at intermediate levels of motility. Why do the cross-feeder never survive if secretions are not essential ? Inside the spatial structures, the density of cells can be derived from (4) and is equal to 0.95 for producers and cross-feeders, and to 0.5 for defectors. This means that the density inside the clusters of cross-feeders and producers is always above the threshold, which means that both strains move according to the reduced motility rate $\epsilon_{\min}$. This motility rate is very low, and has been shown to favor the producers over cross-feeders (Figure 4a). Additionally, if the threshold is above 0.5, the defectors move with the reduced motility rate only if they are near a structure of productive strains, which facilitates exploitation. All in all, if the secretions are not essential, the cross-feeders move according to a exchange rate $\epsilon_{\min}$ where they are not favored while being actively exploited.

If secretions are essential, density-dependent motility favors producers at low motility rates, and some thresholds can favor cross-feeders at intermediate motility rates. The facilitation of producers can be explained by the facilitation of cluster formation, and by $\epsilon_{\min}$ favoring them, but why are cross-feeders favored by intermediate thresholds only ? According to Figure 6c, their victory comes from the exploitation of producers, similarly to what has been observed in unbiased motility (Figure 4c). But why does this exploitation only take place at intermediate thresholds ? According to the spatial structures (Figure 6b), this is because of a trade-off between the cross-feeders' ability to build a sustainable spatial structure and their ability to exploit producers. If the threshold is too low, cross-feeders cannot move away from their spatial structures and are outcompeted by the producers. On the other hand, if the threshold is too high, the cross-feeders don't slow down enough cannot build a sustainable spatial structure before exploiting the producers.

3.2.4 Passive genotypes hinder inter-genotype cooperation

One assumption of our model is that all genotypes in the system make use of the secretions. However, this assumption is likely to be broken in natural ecosystems as a result of the high diversity of natural microbial communities. Similarly to what has been done in the literature in the context of genetic mixing [9], the effect of these passive genotypes will be studied in this section.

To this end, a new "insulator" genotype is introduced to the 2-secretions model. The insulators mimic bacteria of other ecological niches, and grow slowly but steadily. As a result, their fitness is fixed to 0.1 with both unessential and essential secretions, and selection events implying them never outcompete any bacteria. Since their growth rate is fixed and their effect on the other strains depend on their density, their initial density can be changed.

The outcome of the competition can be seen in Figure 7a. If secretions are not essential (data not shown), insulators do not change qualitatively the outcome of the competition compared to Figures 4a and 5a). If secretions are essential however, insulators disfavor cross-feeder and favor producers at low density. At higher densities of insulators, both the cross-feeders and the producers are disfavored. These results are observed both with unbiased motility and with chemotaxis towards all secretions. Altogether, this retrieves the results obtained in [9], except that producers are also disfavored by high densities of insulators. This difference comes from the fact that in our model, a producer is not self-sufficient and needs to be with other producers to grow.

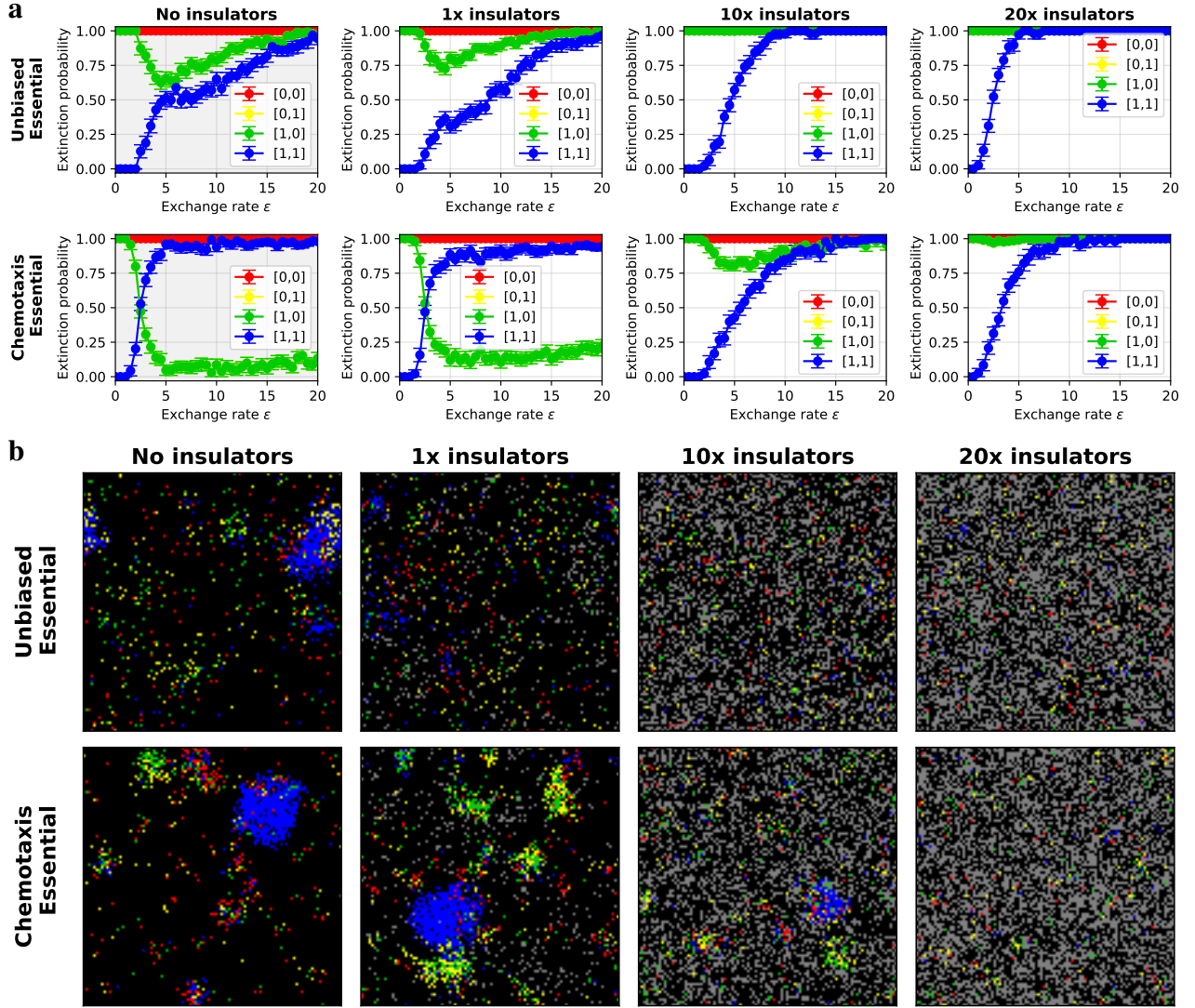


Figure 7: Dynamics of the 2-secretions model with social insulators. **a** Extinction probability of each genotype (except insulators) as a function of the exchange rate, for both unbiased motility and chemotaxis towards all secretions, with essential secretions are for several initial densities of insulators. Initial densities of insulators are expressed in the legends with respect to the initial density of each other genotype. The grayed column depicts data that has been shown before in this report, and which should be taken by the reader as controls for this figure. **b** Examples of spatial structures, for both unbiased motility and chemotaxis towards all secretions, with essential secretions are for several initial densities of insulators. Insulators are displayed with gray pixels. All the snapshots were taken with $\epsilon = 5$ and at generation 20.

The effect of insulators can be seen by looking at the spatial structures (Figure 7b). Their independence on secretions and random motility makes them cover the whole lattice, where they take the space that could be used by the other genotypes to form spatial structures. This decreases the probability that partner cells meet, which harms the cross-feeders the most since their growth requires the meeting of cells of different genotypes. Chemotaxis reduces this effect but does not cancel it, because insulators move randomly and add stochasticity to the other genotypes' motility.

4 Discussion

The 2-secretions model identifies conditions under which microbial genotypes evolve to exchange fitness-promoting secretions. This outcome consistently requires an intermediate motility regime, where there is enough motion to bring partners together, but not enough to let them disperse (Figures 4a, 6a, 7a). This result is similar to what has been observed in [9], where cross-feeding were only selected at intermediate levels of genetic mixing. Other similarities are observed (Figures 2a, 3a): self-sufficiency, which is a property of individuals in [9] and a property of a genotype here, is shown to dominate at low motility and at low genetic mixing [9]. Similarly, exploitation dominates at high values of motility and at high genetic mixing [9]. Altogether, these results suggest that one of the main effects of unbiased motility is to promote mixing between different genotypes, which benefits exploitative genotypes to the cost of self-sufficient genotypes. Interestingly, too much motility can be detrimental for the population, as productive genotypes can go extinct even in the absence of exploitative genotypes (Figures 2d). In this case, extinction is not driven by negative ecological interactions, but by the dispersal of cooperative cells.

Analysis of the genetic mixing model also highlighted the fragility and inefficiency of cross-feeding when it is selected [9]. In this work, this observation is reflected by the factors causing the cross-feeders to dominate, which are often tied to the exploitation of more productive genotypes (Figures 4c, 4d). The Black Queen hypothesis [18] suggests that it may be beneficial for bacteria to lose costly functions that can be obtained through other genotypes because of its leaky nature. The ability of cross-feeders to exploit and outcompete more productive genotypes in the 2-secretions model suggests that the prevalence of cross-feeding in natural ecosystems [2, 4] may come from the loss of one function through Black Queen dynamics, followed by exploitation of producers which still have both functions. This dynamic has also been observed in computational models aiming to understand the evolution of cross-feeders [6]. Here, this exploitation leads to cross-feeders dominating the entire lattice (Figure 4e) and to the extinction of the original producers. However, the resulting population can be expected to be fragile to events like dilutions of the medium, as simulating cross-feeders alone at low density leads to them going extinct (Figure 4d). This needs further examination, as [6] showed that cross-feeding communities were more resilient to nutritional stress.

The constraints mentioned in the previous paragraph are alleviated by chemotaxis towards one or both secretions (Figure 5). Chemotaxis allows to overcome a key constraint of cross-feeding, which is to find a metabolic partner and to stay close to it (Figure 5c). The outcome is however dependent on the essentiality of the secretions, with cross-feeders being broadly selected with essential secretions,

but being dominated by defectors with unessential secretions (Figure 5a). This ties back to real auxotrophies, which are often necessary for growth [2]. Cross-feeding as it was considered in the model is therefore less likely to be selected for compounds which are not necessary for growth, which could take the form of syntrophy in natural ecosystems. Density-dependent motility (Figure 6) also shapes the community differently according to the essentiality of secretions, but only favors cross-feeding with a narrow range of parameters.

Study of the model still display constraints on the evolution of cross-feeding in natural ecosystems. One of these constraints is the low density of cells from other ecological niches, which disfavor cross-feeding by preventing mixing between metabolic partners (Figure 7). However, the fixed growth rate of the insulator cells challenges the application of this result in natural ecosystems, as it causes them to grow homogeneously without forming spatial structures (Figure 7b).

Many aspects of bacterial motility and cross-feeding remain unexplored in the 2-secretions model. For example, motility is assumed to be costless and to have shared properties among genotypes, even though differential motility rates have been shown to affect qualitatively the outcome of the competition with density-dependent motility. Another limit of the model is the number of secretions. In [9], the effect of adding additional secretions is directly linked to their cost, with $[1, 1, 1]$ dominating at lower genetic mixing than $[0, 1, 1]$, which dominates at lower genetic mixing than $[0, 0, 1]$ for example. However, it is unclear what the effects of the increased dependence on other genotypes would be in a cellular automata model where cells have a limited number of neighbors.

References

- [1] Stuart A West, Ashleigh S Griffin, Andy Gardner, and Stephen P Diggle. Social evolution theory for microorganisms. *Nature reviews microbiology*, 4(8):597–607, 2006.
- [2] Glen D’Souza, Shraddha Shitut, Daniel Preussger, Ghada Yousif, Silvio Waschina, and Christian Kost. Ecology and evolution of metabolic cross-feeding interactions in bacteria. *Natural product reports*, 35(5):455–488, 2018.
- [3] Stuart A West, Stephen P Diggle, Angus Buckling, Andy Gardner, and Ashleigh S Griffin. The social lives of microbes. *Annu. Rev. Ecol. Evol. Syst.*, 38(1):53–77, 2007.
- [4] Karoline Faust and Jeroen Raes. Microbial interactions: from networks to models. *Nature Reviews Microbiology*, 10(8):538–550, 2012.
- [5] Sylvie Estrela, J Jeffrey Morris, and Benjamin Kerr. Private benefits and metabolic conflicts shape the emergence of microbial interdependencies. *Environmental microbiology*, 18(5):1415–1427, 2016.
- [6] Miaoxiao Wang, Xiaonan Liu, Yong Nie, and Xiao-Lei Wu. Selfishness driving reductive evolution shapes interdependent patterns in spatially structured microbial communities. *The ISME Journal*, 15(5):1387–1401, 2021.
- [7] Navish Wadhwa and Howard C Berg. Bacterial motility: machinery and mechanisms. *Nature reviews microbiology*, 20(3):161–173, 2022.
- [8] Justin R Seymour, Douglas R Brumley, Roman Stocker, and Jean-Baptiste Raina. Swimming towards each other: the role of chemotaxis in bacterial interactions. *Trends in microbiology*, 32(7):640–649, 2024.
- [9] Nuno M Oliveira, Rene Niehus, and Kevin R Foster. Evolutionary limits to cooperation in microbial communities. *Proceedings of the National Academy of Sciences*, 111(50):17941–17946, 2014.
- [10] Rick Durrett and Simon Levin. Spatial aspects of interspecific competition. *Theoretical population biology*, 53(1):30–43, 1998.
- [11] Tobias Reichenbach, Mauro Mobilia, and Erwin Frey. Mobility promotes and jeopardizes biodiversity in rock–paper–scissors games. *Nature*, 448(7157):1046–1049, 2007.
- [12] Vaishnavi Warriar, Yilin Chen, Ethan Rappaport, Shin Haruta, Hyun Youk, and Babak Momeni. Interplay of spatial structure and interactions in microbial communities. *Environmental Microbiology*, 28(3):e70262, 2026.
- [13] Parker Smith and Martin Schuster. Public goods and cheating in microbes. *Current biology*, 29(11):R442–R447, 2019.

- [14] Daniel T Gillespie. A general method for numerically simulating the stochastic time evolution of coupled chemical reactions. *Journal of computational physics*, 22(4):403–434, 1976.
- [15] Ashleigh S Griffin, Stuart A West, and Angus Buckling. Cooperation and competition in pathogenic bacteria. *Nature*, 430(7003):1024–1027, 2004.
- [16] Rolf Kümmerli, Ashleigh S Griffin, Stuart A West, Angus Buckling, and Freya Harrison. Viscous medium promotes cooperation in the pathogenic bacterium *pseudomonas aeruginosa*. *Proceedings of the Royal Society B: Biological Sciences*, 276(1672):3531, 2009.
- [17] Julien Tailleur and Michael E Cates. Statistical mechanics of interacting run-and-tumble bacteria. *Physical review letters*, 100(21):218103, 2008.
- [18] J Jeffrey Morris, Richard E Lenski, and Erik R Zinser. The black queen hypothesis: evolution of dependencies through adaptive gene loss. *MBio*, 3(2):10–1128, 2012.

Annex

Algorithm 1 Selection event

Require: Position (x, y) on the lattice occupied by a bacteria.

Chose (x', y') a randomly an adjacent position among $(x + 1, y)$, $(x - 1, y)$, $(x, y + 1)$ or $(x, y - 1)$.

if Position x' is also occupied **then**

if Both bacteria are not of the same strain **then** $\triangleright$ No selection between $[0, 1]$ and $[1, 0]$

if $f_{(x,y)} < f_{(x',y')}$ **then**

 Kill the bacteria at position (x, y) .

else if $f_{(x,y)} = f_{(x',y')}$ **then**

 Kill the bacteria at position (x, y) with probability $\frac{1}{2}$.

end if

end if

end if

Algorithm 2 Reproduction event

Require: Position (x, y) on the lattice occupied by a bacteria, fitness $f_{(x,y)}$ of this bacteria.

Chose (x', y') a randomly an adjacent position among $\{(x + 1, y), (x - 1, y), (x, y + 1), (x, y - 1)\}$.

if Position (x', y') is not occupied **then**

 Draw $X \sim \text{Unif}([0, 1])$

if $f_{(x,y)} > X$ **then**

 Copy the cell type of position (x, y) into position (x', y') .

end if

end if

Algorithm 3 Death event

Require: Position (x, y) on the lattice occupied by a bacteria.

Kill the bacteria at position (x, y) .

Algorithm 4 Stopping condition

Require: Lattice size L

while Cells of two or more strains are present on the lattice **do** $\triangleright \{[0, 0], ([0, 1], [1, 0]), [1, 1]\}$

 Simulate one generation

end while

for $i \in \llbracket 1, L \rrbracket$ **do**

 Simulate one generation

end for

if There are more than $L/4$ cells alive **then**

 Label the remaining genotypes as surviving, and the genotypes without living cells as extinct

else

 Label all genotypes as extinct

end if

Algorithm 5 Initialization in the 2-secretions model

Require: Size L of the lattice, initial sparsity of cells S .

```
for  $i \in \llbracket 1, L \rrbracket$  do
  for  $j \in \llbracket 1, L \rrbracket$  do
    Draw  $X \sim \text{Unif}(\llbracket 1, S \rrbracket)$ 
    if  $X = 1$  then
      Put  $[0, 0]$  at space  $(i, j)$ 
    else if  $X = 2$  then
      Put  $[0, 1]$  at space  $(i, j)$ 
    else if  $X = 3$  then
      Put  $[1, 0]$  at space  $(i, j)$ 
    else if  $X = 4$  then
      Put  $[1, 1]$  at space  $(i, j)$ 
    else
      Put an empty space at  $(i, j)$ 
    end if
  end for
end for
```

Algorithm 6 Chemotaxis

Require: Position (x, y) on the lattice occupied by a bacteria. Genotype $G \in \llbracket 0, 3 \rrbracket$ of the bacteria, where $\llbracket 0, 3 \rrbracket$ is assimilated with $\{[0, 0], [0, 1], [1, 0], [1, 1]\}$, detection radius R

```
for  $(x', y')$  in  $\{(x+1, y), (x-1, y), (x, y+1), (x, y-1)\}$  do
   $C \leftarrow (0, 0, 0, 0)$   $\triangleright$  Count cells of each genotype around the adjacent position
  for  $i$  in  $\llbracket x' - R, x' + R \rrbracket$  do
    for  $j$  in  $\llbracket y' - R, y' + R \rrbracket$  do
      if Position  $(i, j)$  is occupied then
         $g \leftarrow$  genotype of cell at position  $(i, j)$ 
         $C[g] \leftarrow c[g] + 1$ 
      end if
    end for
  end for
   $w_{(x', y')} \leftarrow C \cdot (0, 1, 1, 1)$   $\triangleright$  Dot product to get the number of secretors
end for
Chose  $(x', y') \in \{(x+1, y), (x-1, y), (x, y+1), (x, y-1)\}$  with probability  $w_{(x', y')}/\sum w$ .
```

Algorithm 7 Density-dependant motility

Require: Position (x, y) on the lattice occupied by a bacteria, exchange rate ε , reduced exchange rate $\varepsilon_{\min}$, detection radius R , density-dependent motility threshold T .

$D \leftarrow 0$

▷ Compute local density of cells

```
for  $i$  in  $\llbracket x - R, x + R \rrbracket$  do
  for  $j$  in  $\llbracket y - R, y + R \rrbracket$  do
    if Position  $(i, j)$  is occupied then
       $D \leftarrow D + 1$ 
    end if
  end for
end for
 $d \leftarrow D / (2R + 1)^2$ 
if  $d > T$  then
  Move with probability  $\varepsilon_{\min} / \varepsilon$ .
else
  Move.
end if
```

Table 1: Usual parameter values.

Parameter name	Notation	Usual value
Lattice size	L	100
Selection rate	σ	1
Reproduction rate	μ	1
Death rate	δ	0.05
Exchange rate	ε	$\varepsilon \in [0, 20]$
Fitness benefit of secretions	g	0.1
Fitness cost of secretions	c	0.05
Basal fitness	b	0.1 (unessential) or 0 (essential)
Weight in the fitness formula	ω	0.5 (unessential) or 0 (essential)
Radius of secretions	r	2
Initial sparsity of cells	S	50
Detection radius	R	3
Reduced exchange rate	$\varepsilon_{\min}$	0.1 (with density-dependent motility)
Density threshold	T	0.5
Initial density of insulators	I	0

Parameters values used to get the data for the figures, when their caption does not say otherwise.

Résumé

Les bactéries échangent souvent des métabolites avec des bactéries d'autres génotypes, via la diffusion de ces métabolites dans le milieu. Ces relations de cross-feeding peuvent concerner des composés coûteux et qui peuvent être essentiels pour la croissance bactérienne comme des enzymes extracellulaires, des molécules de récupération, des sources de carbone, des agents de détoxification ou des acides aminés. Cette dépendance sur d'autres génotypes peut être néfaste, puisque les partenaires métaboliques ne peuvent pas se reproduire s'ils ne sont pas proches. Dans ce rapport, la motilité est examinée comme une réponse à ce problème via un modèle de compétition entre des partenaires cross-feeders, un génotype autosuffisant qui produit tous les composés et un génotype exploiteur qui a besoin des composés pour se reproduire mais qui n'en produit aucun. L'analyse de ce modèle montre que le cross-feeding n'est sélectionné qu'à des niveaux intermédiaires de motilité, ce qui permet aux partenaires métaboliques de se rencontrer sans les laisser se disperser. Cependant, cette issue n'est causée que par l'exploitation de génotypes plus productifs si les sécrétions sont essentielles à la croissance. Cette contrainte sur la motilité est fortement réduite par la chimiotaxie avec des sécrétions essentielles, qui permet aux partenaires de se rencontrer avec plus d'efficacité et de rester ensemble. La motilité densité-dépendente, où les bactéries ralentissent en réponse à une forte densité cellulaire locale, ne favorise le cross-feeding que dans des cas précis caractérisés par le seuil de densité. Cependant, le cross-feeding requiert de la coopération locale, qui peut être défavorisée dans des écosystèmes naturels à cause de cellules d'autres niches écologiques, qui prennent de la place sans produire ou requérir de sécrétions.

Abstract

Bacteria often exchange metabolites with bacteria of different genotypes, through the diffusion of those metabolites in the medium. These cross-feeding interactions can concern costly compounds which can be essential for bacterial growth such as extracellular enzymes, scavenging molecules, carbon sources, detoxification agents or amino acids. This reliance on other genotypes can be detrimental, as metabolic partners cannot grow if they are not in close proximity. Here, motility is examined as a response to this problem in a competition model between cross-feeding partners, a self-sufficient genotype which produces all compounds and an exploitative genotype which needs the compounds to grow but does not produce any. Analysis of this model shows that cross-feeding is only selected at intermediate levels of motility, which allows the metabolic partners to meet without letting them disperse. However, this outcome is only due to exploitation of more productive genotypes when the secretions are essential for growth. This motility constraint is strongly mitigated by chemotaxis with essential secretions, which allows the partners to meet with more efficiency and to stay together. Density-dependent motility, where bacteria slow down in response to a high local density of cells, is only shown to favor cross-feeding in precise cases characterized by the density threshold. However, cross-feeding requires local cooperation, which can be disfavored in natural ecosystems because of cells from other ecological niches taking up space without producing or needing the secretions.